Fe-C Problem 4 | Problems T4_PhaseDiagrams II.pdf #4 | Fe-C · ledeburite

Phases and Microconstituents of a Eutectic Cast Iron

Original statement · Problems T4_PhaseDiagrams II.pdf, p. 2

4.- For an alloy Fe-4.3% of C, at the following temperatures: T_1 = 1147 + ΔT ºC; T_2 = 1147 – ΔT ºC; T_3 = 900 ºC; T_4 = 727 + ΔT ºC; T_5 = 727 – ΔT ºC. a) Indicate the phases present, their chemical composition and their mass fractions. b) Indicate the micro constituents, their chemical composition and their mass fractions.

Official Solution: a) T_1: C_L = 4.3% C, w_L = 1; T_2: C_{Fe_3C} = 6.7% C, C_γ = 2.14% C, w_γ = 0.53; T_3: C_γ = 1.3% C, C_{Fe_3C} = 6.7% C, w_γ = 0.44; T_4: C_γ = 0.76% C, C_{Fe_3C} = 6.7% C, w_γ = 0.40; T_5: C_α = 0.022% C, C_{Fe_3C} = 6.7% C, w_α = 0.36; b) At T_2: $w_{Fe_3C,E}$ 0.47, $w_{\gamma,E}$ 0.53

PHASE 1 · HYPOTHESES & PARAMETERS

1. Equilibrium cooling, Fe-Fe3C diagram [Session 9 Slide 4]; $C_0 = 4.3$ % C is the **eutectic composition**: the alloy is liquid down to 1147 °C and then becomes 100 % **ledeburite**, $L \rightarrow \gamma\,(2.14) + \mathrm{Fe_3C}\,(6.67)$ [Session 9 Slide 10].
2. **Reading of the figure:** A_{cm} at 900 °C ≈ 1.3 % C (as in Problem FC03).

PHASE 2 · FRAMES & STRATEGY

Frame: the Fe-Fe3C diagram; one tie line per temperature. Below 1147 °C the alloy stays in γ + Fe3C, and the γ of the ledeburite loses carbon along A_{cm} (more cementite precipitates), until at 727 °C it becomes pearlite ("transformed ledeburite", Session 9 Slide 11).

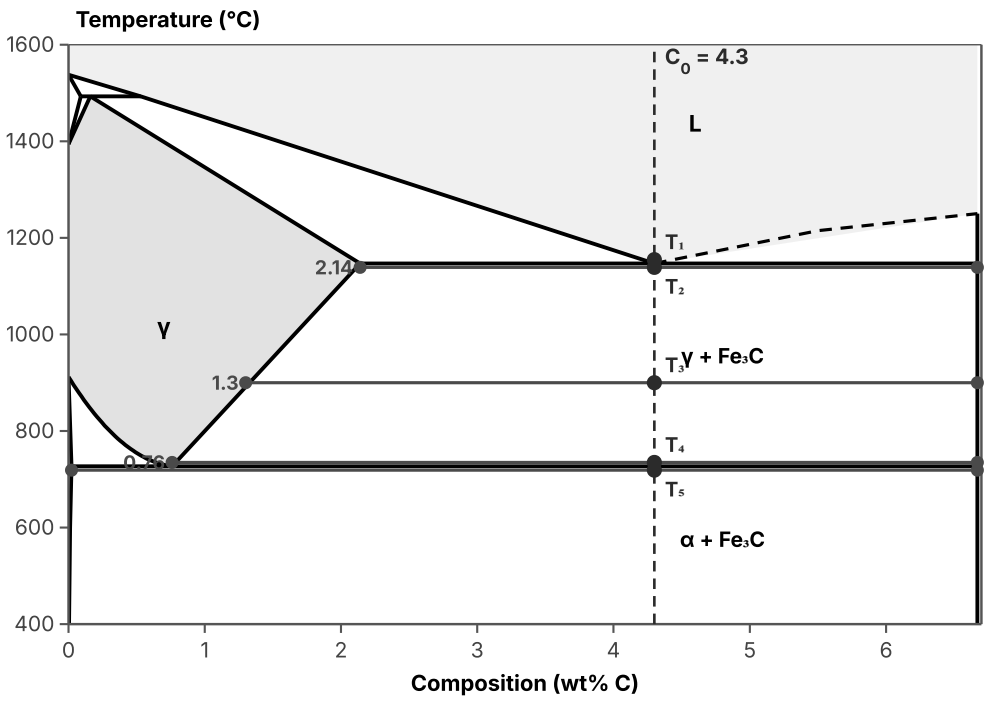


Figure 32. Fe-4.3 % C: liquid above 1147 °C, ledeburite (γ + Fe_3C) below it, and the tie lines at 900 °C and 727 ± ΔT.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Phases T_1: liquid, 4.3 % C, $W_L = 1$.

T_2 (γ 2.14 / Fe3C 6.67; length 4.53):

$$W_\gamma = \frac{6.67 - 4.3}{4.53} = \frac{2.37}{4.53} = 0.523 \qquad W_{Fe_3C} = \frac{4.3 - 2.14}{4.53} = \frac{2.16}{4.53} = 0.477$$

T_3 (γ 1.3 / Fe3C 6.67; length 5.37):

$$W_\gamma = \frac{6.67 - 4.3}{5.37} = \frac{2.37}{5.37} = 0.441 \qquad W_{Fe_3C} = 0.559$$

T_4 (γ 0.76 / Fe3C 6.67; length 5.91):

$$W_\gamma = \frac{2.37}{5.91} = 0.401 \qquad W_{\mathrm{Fe_3C}} = 0.599$$

T_5 (α 0.022 / Fe3C 6.67; length 6.648):

$$W_\alpha = \frac{2.37}{6.648} = 0.356 \qquad W_{\mathrm{Fe_3C}} = 0.644$$

b) Microconstituents

- T_1: liquid.
- T_2: 100 % **ledeburite** (4.3 % C overall), made of $W_{\gamma,E} = 0.523$ and $W_{\mathrm{Fe_3C},E} = 0.477$ (Worked example 14 of the theory).
- T_3, T_4: ledeburite whose austenite has rejected cementite: γ + Fe3C in the amounts of part (a).
- T_5: **transformed ledeburite**: the 0.401 of austenite at T_4 has become pearlite, inside a cementite matrix (0.599).

PHASE 4 · CHECKS & INTERPRETATION

- Mass balance at T_2: $0.523 \times 2.14 + 0.477 \times 6.67 = 1.12 + 3.18 = 4.30$ % C .
- The amount of cementite only grows on cooling (0.477 → 0.559 → 0.599 → 0.644), because the solubility of C in the iron phase only falls.

Check against the official solution

Agrees with the key at T_3, T_4, T_5 and in part (b).

> DISCREPANCY WITH THE OFFICIAL SOLUTION
>
> At T_2 the key gives $w_\gamma = 0.53$. With 6.67 % C for cementite (the value of the slides), $w_\gamma = 2.37/4.53 = 0.523$ (0.52); the key's 0.53 follows from 6.7 % C, $2.4/4.56 = 0.526$. The difference is only the rounding of the cementite composition; 0.523 is kept.